

Assignment – 4.3

Q - 1. List 5 common BIOS or POST error messages you might see when starting a PC and write what hardware or configuration issue each one usually points to.

No.	Error Message	Problem
1.	CMOS Checksum Error	CMOS battery or BIOS settings problem
2.	Keyboard Error	Keyboard is disconnected or faulty
3.	No Boot Device Found	HDD/SSD is not detected or boot order is wrong
4.	Memory Error	RAM is loose, faulty, or not detected
5.	CPU Fan Error	CPU fan is disconnected or not working properly

Q - 2. Given this scenario: On boot, a PC beeps 3 times in a repeating pattern and does not display anything on the screen. Research and write what this POST beep code typically means for an AMI BIOS, and suggest two troubleshooting steps.

AMI BIOS – 3 Beeps

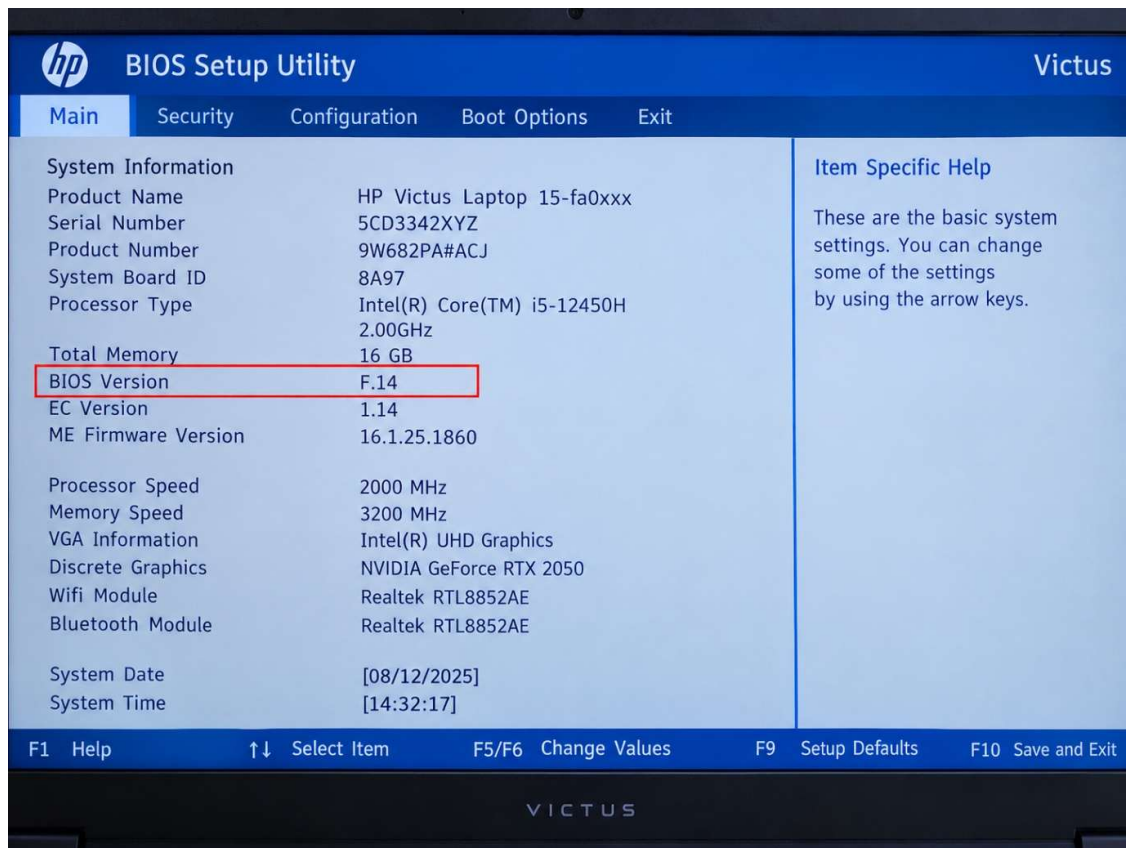
Problem:

➤ 3 beeps usually mean **RAM (memory) error**.

Troubleshooting:

1. Remove and properly **reseat the RAM**.
2. Try **one RAM stick at a time** or replace faulty RAM.

Q - 3. Access the BIOS setup utility on any available PC or virtual machine, take a screenshot of the main BIOS screen, and identify the motherboard manufacturer and BIOS version.



➤ **Motherboard Manufacturer:** HP

➤ **BIOS Version:** F.14

Q - 4. You are given a screenshot of a POST error message that says 'CMOS checksum error – Defaults loaded.' Explain what this means and describe how you would resolve this issue in a real system. Hint: Think about the CMOS battery and BIOS settings reset.

CMOS Checksum Error – Defaults Loaded

Meaning:

- BIOS settings are lost or incorrect.
- The computer loads the default BIOS settings.
- Usually caused by a **weak or dead CMOS battery**.

Solution:

1. Restart the computer and open **BIOS Setup**.
2. Select **Load Default Settings**.
3. Set the correct **date and time**.
4. Save and restart.
5. If the error comes back, **replace the CMOS battery (CR2032)**.
6. Set the BIOS settings again.